

Autoencoders

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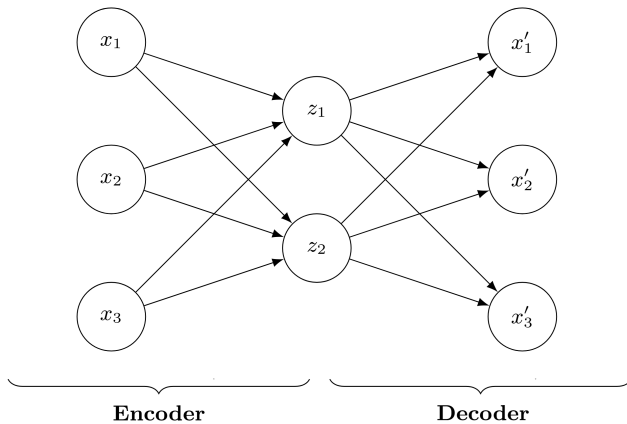
Insper

A rede neural denominada *autoencoder* tenta produzir na camada de saída os mesmos valores da camada de entrada.

Em outras palavras, o propósito de um autoencoder é aprender a função identidade entre as camadas de entrada e de saída.

Ao fazer isto, o autoencoder consegue aprender uma *representação latente* dos dados (por exemplo, de dimensão mais baixa) em suas camadas internas.

A aprendizagem dos parâmetros do autoencoder é feita via *backpropagation*, como fizemos com as diversas arquiteturas das aulas anteriores.

Autoencoder

$$s_1 = v_{11}x_1 + v_{12}x_2 + v_{13}x_3 + a_1$$

$$s_2 = v_{21}x_1 + v_{22}x_2 + v_{23}x_3 + a_2$$

$$z_1 = \sigma(s_1)$$

$$z_2 = \sigma(s_2)$$

$$t_1 = w_{11}z_1 + w_{12}z_2 + b_1$$

$$t_2 = w_{21}z_1 + w_{22}z_2 + b_2$$

$$t_3 = w_{31}z_1 + w_{32}z_2 + b_3$$

$$x'_1 = \sigma(t_1)$$

$$x'_2 = \sigma(t_2)$$

$$x'_3 = \sigma(t_3)$$

$$L = \frac{1}{2} \left((x'_1 - x_1)^2 + (x'_2 - x_2)^2 + (x'_3 - x_3)^2 \right)$$

$$\frac{\partial L}{\partial a_2} = \left(\frac{\partial L}{\partial x'_1} \frac{\partial x'_1}{\partial t_1} \frac{\partial t_1}{\partial z_2} + \frac{\partial L}{\partial x'_2} \frac{\partial x'_2}{\partial t_2} \frac{\partial t_2}{\partial z_2} + \frac{\partial L}{\partial x'_3} \frac{\partial x'_3}{\partial t_3} \frac{\partial t_3}{\partial z_2} \right) \frac{\partial z_2}{\partial s_2} \frac{\partial s_2}{\partial a_2}$$

$$\frac{\partial L}{\partial v_{11}} = ((x'_1 - x_1)\sigma'(t_1)w_{11} + (x'_2 - x_2)\sigma'(t_2)w_{21} + (x'_3 - x_3)\sigma'(t_3)w_{31})\sigma'(s_1)x_1$$

$$\frac{\partial L}{\partial v_{12}} = ((x'_1 - x_1)\sigma'(t_1)w_{11} + (x'_2 - x_2)\sigma'(t_2)w_{21} + (x'_3 - x_3)\sigma'(t_3)w_{31})\sigma'(s_1)x_2$$

$$\frac{\partial L}{\partial v_{13}} = ((x'_1 - x_1)\sigma'(t_1)w_{11} + (x'_2 - x_2)\sigma'(t_2)w_{21} + (x'_3 - x_3)\sigma'(t_3)w_{31})\sigma'(s_1)x_3$$

$$\frac{\partial L}{\partial a_1} = ((x'_1 - x_1)\sigma'(t_1)w_{11} + (x'_2 - x_2)\sigma'(t_2)w_{21} + (x'_3 - x_3)\sigma'(t_3)w_{31})\sigma'(s_1)$$

$$\frac{\partial L}{\partial v_{21}} = ((x'_1 - x_1)\sigma'(t_1)w_{12} + (x'_2 - x_2)\sigma'(t_2)w_{22} + (x'_3 - x_3)\sigma'(t_3)w_{32})\sigma'(s_2)x_1$$

$$\frac{\partial L}{\partial v_{22}} = ((x'_1 - x_1)\sigma'(t_1)w_{12} + (x'_2 - x_2)\sigma'(t_2)w_{22} + (x'_3 - x_3)\sigma'(t_3)w_{32})\sigma'(s_2)x_2$$

$$\frac{\partial L}{\partial v_{23}} = ((x'_1 - x_1)\sigma'(t_1)w_{12} + (x'_2 - x_2)\sigma'(t_2)w_{22} + (x'_3 - x_3)\sigma'(t_3)w_{32})\sigma'(s_2)x_3$$

$$\frac{\partial L}{\partial a_2} = ((x'_1 - x_1)\sigma'(t_1)w_{12} + (x'_2 - x_2)\sigma'(t_2)w_{22} + (x'_3 - x_3)\sigma'(t_3)w_{32})\sigma'(s_2)$$

$$\frac{\partial L}{\partial w_{11}} = \frac{\partial L}{\partial x'_1} \frac{\partial x'_1}{\partial t_1} \frac{\partial t_1}{\partial w_{11}}$$

$$\frac{\partial L}{\partial w_{12}} = \frac{\partial L}{\partial x'_1} \frac{\partial x'_1}{\partial t_1} \frac{\partial t_1}{\partial w_{12}}$$

$$\frac{\partial L}{\partial w_{21}} = \frac{\partial L}{\partial x'_2} \frac{\partial x'_2}{\partial t_2} \frac{\partial t_2}{\partial w_{21}}$$

$$\frac{\partial L}{\partial w_{22}} = \frac{\partial L}{\partial x'_2} \frac{\partial x'_2}{\partial t_2} \frac{\partial t_2}{\partial w_{22}}$$

$$\frac{\partial L}{\partial w_{31}} = \frac{\partial L}{\partial x'_3} \frac{\partial x'_3}{\partial t_3} \frac{\partial t_3}{\partial w_{31}}$$

$$\frac{\partial L}{\partial w_{32}} = \frac{\partial L}{\partial x'_3} \frac{\partial x'_3}{\partial t_3} \frac{\partial t_3}{\partial w_{32}}$$

$$\frac{\partial L}{\partial b_1} = \frac{\partial L}{\partial x'_1} \frac{\partial x'_1}{\partial t_1} \frac{\partial t_1}{\partial b_1}$$

$$\frac{\partial L}{\partial b_2} = \frac{\partial L}{\partial x'_2} \frac{\partial x'_2}{\partial t_2} \frac{\partial t_2}{\partial b_2}$$

$$\frac{\partial L}{\partial b_3} = \frac{\partial L}{\partial x'_3} \frac{\partial x'_3}{\partial t_3} \frac{\partial t_3}{\partial b_3}$$

$$\frac{\partial L}{\partial w_{11}} = (x'_1 - x_1) \sigma'(t_1) z_1$$

$$\frac{\partial L}{\partial w_{12}} = (x'_1 - x_1) \sigma'(t_1) z_2$$

$$\frac{\partial L}{\partial w_{21}} = (x'_2 - x_2) \sigma'(t_2) z_1$$

$$\frac{\partial L}{\partial w_{22}} = (x'_2 - x_2) \sigma'(t_2) z_2$$

$$\frac{\partial L}{\partial w_{31}} = (x'_3 - x_3) \sigma'(t_3) z_1$$

$$\frac{\partial L}{\partial w_{32}} = (x'_3 - x_3) \sigma'(t_3) z_2$$

$$\frac{\partial L}{\partial b_1} = (x'_1 - x_1) \sigma'(t_1)$$

$$\frac{\partial L}{\partial b_2} = (x'_2 - x_2) \sigma'(t_2)$$

$$\frac{\partial L}{\partial b_3} = (x'_3 - x_3) \sigma'(t_3)$$